DRUGS · ANTIBIOTICS: OTHER CLASSES

Linezolid

Linezolid is an oxazolidinone active against Gram-positive bacteria, including MRSA and vancomycin-resistant enterococci, and it is almost fully absorbed by mouth. In ENT it is used, on microbiology advice, for MRSA infections of the neck, flaps and wounds when glycopeptides cannot be used or oral treatment is needed. Courses longer than two weeks cause myelosuppression, and it can trigger serotonin syndrome.

1 CHAPTER ONE At a glance

The drug on one page.

1.1 Summary

LINEZOLID

	SUMMARY
Class	Oxazolidinone
How it works	Binds the 23S rRNA of the 50S subunit and prevents formation of the initiation complex; bacteriostatic for staphylococci
Spectrum	MRSA, coagulase-negative staphylococci, streptococci including resistant pneumococci, enterococci including VRE, Nocardia, some mycobacteria; no Gram-negative activity
ENT uses	MRSA soft tissue, wound and neck infection; resistant Gram-positive infection needing oral treatment; part of regimens for drug-resistant tuberculosis
Adult dose	600 mg twice daily orally or intravenously
Child dose	Under 12 years 10 mg/kg every 8 hours (maximum 600 mg); 12 years and over as adult
Duration	Usually up to 14 days without extra monitoring; longer courses need weekly blood counts

	SUMMARY
Avoid	Within 2 weeks of an MAOI; uncontrolled hypertension, phaeochromocytoma, thyrotoxicosis unless monitored
Pregnancy	Avoid unless essential

2 CHAPTER TWO How it works

Mechanism and handling.

2.1 Mechanism

A unique ribosomal site means no cross-resistance with other classes. It reduces toxin production, an advantage in toxin-mediated staphylococcal and streptococcal infection.

2.2 Handling

About 100% oral bioavailability, so step-down from intravenous to oral therapy can happen early. Good soft tissue and bone penetration. Non-renal metabolism.

3 CHAPTER THREE Safety

What goes wrong.

3.1 Adverse effects

- **Myelosuppression** (thrombocytopenia, anaemia) with courses beyond 2 weeks: weekly full blood count.
- **Peripheral and optic neuropathy** with courses over 28 days; check vision.
- Lactic acidosis; hypoglycaemia in diabetics; C. difficile.

3.2 Interactions

A weak, reversible **MAO inhibitor**: serotonin syndrome with SSRIs, SNRIs, tricyclics, triptans and pethidine; hypertensive reactions with sympathomimetics such as pseudoephedrine and large amounts of tyramine-rich food.

CHAPTER FOUR

4 For the examination

What the examiner wants to hear.

4.1 Viva essentials

WHAT THE EXAMINER IS LOOKING FOR

- Its role in **MRSA** and VRE, and the advantage of oral bioavailability.
- Monitoring: blood count, vision, lactate.
- The **serotonin syndrome** and pseudoephedrine interactions.

4.2 Common pitfalls

COMMON PITFALLS

- Starting linezolid in a patient on an SSRI without discussion and monitoring.
- Prolonged courses without blood counts.

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